

1. Fill in the blank spaces. Justifications are not required for this question.

(a) (1 point) The exponential form of $1 - i$ is $\sqrt{2}e^{-i\pi/4}$.

(b) (1 point) The cartesian form of $8e^{7i\pi/6}$ is $-4\sqrt{3} - 4i$.

(c) (1 point) The degree of the polynomial $x^{11} + x^2 - x$ is 11 .

(d) (1 point) Consider the polynomial $P(x) = x^4(x + 3)^5(x - 3)^6$.

i. (1 point) The degree of $P(x)$ is 15 .

ii. (1 point) The root -3 is of multiplicity 5 .

iii. (1 point) 0 is a root of multiplicity 4 .

2. (5 points) Find all solutions to the equation

$$z^3 = 8e^{i\pi/6}.$$

Write the solutions in exponential form using the principal value of the argument.

Solution: If $z = re^{i\theta}$ is a solution of the equation, then

$$(re^{i\theta})^3 = 2^3e^{i\pi/6}.$$

Hence $r = 2$ and

$$\begin{aligned} 3\theta &= \pi/6 + 2\pi k = \frac{(1 + 12k)\pi}{6} \\ \Leftrightarrow \theta &= \frac{(1 + 12k)\pi}{18}. \end{aligned}$$

for some integer k . For $k = 0, 1, 2$ we find the three angles and solutions

$$\begin{array}{lll} k = 0 & \theta = \pi/18 & z_0 = 2e^{i\pi/18} \\ k = 1 & \theta = 13\pi/18 & z_1 = 2e^{13i\pi/18} \\ k = 2 & \theta = 25\pi/18 & z_2 = 2e^{25i\pi/18} = 2e^{-11i\pi/18}. \end{array}$$

Hence, the three solutions that we were looking for are

$$2e^{i\pi/18}, \quad 2e^{13i\pi/18} \quad \text{and} \quad 2e^{-11i\pi/18},$$

where the three angles are the principal values of the argument.

3. Consider the polynomial $P(x) = 3x^5 - 2x^4 + 3x^3 - 5x^2 + 5x - 7$.

(a) (2 points) Use Bounds theorem to find bounds on the roots.

Solution: By Bounds theorem, if x is a root of $P(x)$ then

$$|x| < \frac{M}{3} + 1 = \frac{7}{3} + 1 = \frac{10}{3} = 3 + \frac{1}{3} = 3.333333\dots$$

where $M = \max\{|-2|, |3|, |-5|, |5|, |-7|\} = 7$.

(b) (3 points) Use Descartes rule of signs to determine the number of possible positive and negative real roots.

Solution: There are 5 sign changes in the coefficients of $P(x)$, so there are 1, 3 or 5 positive real roots. The polynomial

$$P(-x) = -3x^5 - 2x^4 - 3x^3 - 5x^2 - 5x - 7$$

has 0 sign changes so $P(x)$ has no negative real roots.

(c) (3 points) Use parts (a) and (b) to deduce that all the roots of the polynomial are contained in the interval $(0, 7)$.

Solution: By part (a), we know that all the roots are contained in the interval $(-10/3, 10/3)$. On the other hand, by part (b) we know that $P(x)$ has no negative roots, in particular it has no roots in the interval $(-10/3, 0)$. From these two facts, it follows that all the roots must be in the interval $[0, 10/3)$. Moreover, 0 is not a root since $P(0) = -7$. Hence, all the roots are in fact in the interval $(0, 10/3)$. Finally, since $10/3 = 3.333\dots < 7$, the interval $(0, 7)$ contains the interval $(0, 10/3)$. We deduce that all the roots are also $(0, 7)$.

(d) (4 points) Use the Rational root theorem to give a list of possible rational roots. Reduce the list as much as possible by using the previous parts.

Solution: By Rational root theorem, if p/q is a root in reduced form then p divides -7 and q divides 3. So the possible values for p are $\pm 1, \pm 7$, the possible values for q are $\pm 1, \pm 3$ and the possible values for p/q are $\pm 1, \pm 7, \pm 1/3, \pm 7/3$. By part (a) we can remove all the roots of absolute value greater than $10/3$. Since

$$|\pm 7| = 7 > \frac{10}{3} = 3.3333\dots$$

we can remove ± 7 from the list. By part (b) we can also remove the negative roots. Thus, the possible rational roots are

$$1, \quad \frac{1}{3}, \quad \text{and} \quad \frac{7}{3}.$$

4. (5 points) Consider the polynomial $P(x) = 2x^{77} - m + 1$ where m is a real number. Find all values of m such that $(x + 1)$ divides $P(x)$. Justify your answer.

Solution: By the remainder theorem, the division of $P(x)$ by $x + 1$ has remainder $r = P(-1)$. Thus, if we want to find all values of m such that $(x + 1)$ divides $P(x)$, we need to find all values of m such

that the remainder is 0. That is we want to find all real numbers m such that

$$\begin{aligned} 0 &= P(-1) \\ &= 2(-1)^{77} - m + 1 \\ &= -2 - m + 1 \\ &= -1 - m. \end{aligned}$$

Hence, the remainder is 0 precisely when $-1 - m = 0$, that is when $m = -1$.

5. (6 points) Consider the polynomial $P(x) = x^3 + x^2 + 4x + 30$. It is given to you that $1 - 3i$ is a root of $P(x)$. List all the roots of $P(x)$.

Solution: The polynomial $P(x)$ has real coefficients. Hence, if $1 - 3i$ is a root, then the complex conjugate $1 + 3i$ is a root as well. In particular,

$$(x - (1 - 3i))(x - (1 + 3i)) = x^2 - 2x + 10$$

is a factor of $P(x)$. We perform polynomial division:

$$\begin{array}{r} x^2 - 2x + 10 \overline{) \begin{array}{r} x^3 + x^2 + 4x + 30 \\ - x^3 + 2x^2 - 10x \\ \hline 3x^2 - 6x + 30 \\ - 3x^2 + 6x - 30 \\ \hline 0 \end{array}} \end{array}$$

and find that

$$P(x) = (x^2 - 2x + 10)(x + 3).$$

Hence, the roots of $P(x)$ are

$$1 - 3i, \quad 1 + 3i, \quad \text{and} \quad -3.$$